

MARK HANEY

PYTHON BACKEND ENGINEER | STAFF SOFTWARE ENGINEER | FASTAPI | DJANGO | AWS | MICROSERVICES | CI/CD

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PROFESSIONAL SUMMARY

Senior Python Backend Engineer with 12+ years of experience across SaaS, property technology, nonprofit software, civic technology, and IT services, specializing in **Python 3.8-3.12**, **FastAPI 0.95-0.110**, **Django 3.2-5.0**, **PostgreSQL 12-16**, **AWS**, and microservice platforms that support secure API delivery, workflow automation, reporting, payments, tenant operations, and high-volume data processing. Strong background in system design, cloud-native architecture, CI/CD, observability, database optimization, and production incident resolution, with a record of improving API latency, deployment reliability, data accuracy, and engineering velocity through pragmatic backend ownership.

TECH SKILLS

- ❖ **Backend:** Python 3.8-3.12, FastAPI 0.95-0.110, Django 3.2-5.0, Flask 1.1-3.0, Celery 4-5, Pydantic 1.10-2.x, SQLAlchemy 1.4-2.0, REST APIs, GraphQL APIs, Microservices
- ❖ **Databases:** PostgreSQL 12-16, MongoDB 4-7, MySQL 8, Redis 5-7, Elasticsearch 7-8, database indexing, query tuning, schema migration
- ❖ **Cloud & DevOps:** AWS, ECS, Lambda, S3, RDS, CloudWatch, Docker 20-24, Kubernetes 1.22-1.29, Terraform 1.x, GitHub Actions, Jenkins
- ❖ **Testing & Quality:** Pytest 6-8, unittest, integration testing, contract testing, API testing, test automation, code review, static analysis
- ❖ **Architecture:** Distributed systems, event-driven architecture, background jobs, message queues, domain-driven design, observability, performance optimization

PROFESSIONAL EXPERIENCE

Staff Software Engineer

Relatio

Jun 2022 – Apr 2026

Las Vegas, NV

- ❖ Led backend architecture for a SaaS workflow and analytics platform using **Python 3.10-3.12**, **FastAPI**, **PostgreSQL**, and **AWS ECS**, improving API response reliability through better service boundaries and async request handling.
- ❖ Designed versioned REST APIs with **FastAPI 0.100+**, **Pydantic 2.x**, and OpenAPI contracts, reducing frontend-backend integration defects by **18%** and helping product teams release features with fewer rollback risks.
- ❖ Migrated legacy synchronous Python services into containerized microservices on **Docker**, **AWS ECS**, and **GitHub Actions**, reducing deployment failure issues through repeatable builds and environment validation.
- ❖ Resolved recurring production errors caused by slow ORM queries, missing composite indexes, and inconsistent pagination logic by optimizing **SQLAlchemy 2.0** models and PostgreSQL execution plans.
- ❖ Implemented background processing with **Celery 5**, Redis queues, and idempotent retry logic to stabilize report generation workflows that previously failed during peak data loads.

- ❖ Created observability standards using structured logging, **CloudWatch**, tracing metadata, and service-level dashboards, improving incident diagnosis speed by **31%** during production support rotations.
- ❖ Refactored authentication and permission checks into reusable backend middleware, strengthening API security while reducing duplicated authorization code across multiple service modules.
- ❖ Improved CI/CD pipelines with automated **Pytest**, Docker image validation, migration checks, and staged deployment gates, increasing release confidence across backend teams.
- ❖ Guided engineers on API design, clean service layers, and database migration strategy, helping the team maintain consistent patterns across new Python services.
- ❖ Solved data synchronization defects between customer-facing modules and internal reporting jobs by introducing transaction-safe update flows and reconciliation checks.
- ❖ Partnered with product managers to implement new workflow automation features, including configurable rules, audit trails, and scheduled processing for enterprise customers.
- ❖ Reviewed architecture decisions for scalability, maintainability, and operational risk, ensuring backend changes supported long-term platform growth instead of short-term patching.

Senior Software Engineer

AppFolio

Feb 2018 – May 2022

Goleta, CA

- ❖ Built backend services for property technology products using **Python 3.8-3.10**, **Django 3.2**, **PostgreSQL**, Redis, and AWS, improving rental workflow processing speed by **24%**.
- ❖ Implemented tenant communication and payment-related API features with secure validation, transaction logging, and background job handling to reduce customer support issues by **16%**.
- ❖ Migrated older Django modules from function-heavy views into service-oriented patterns, improving test coverage and making business rules easier to reuse across applications.
- ❖ Fixed production bugs involving duplicate payment notifications, timeout errors, and inconsistent status transitions by adding idempotency keys and stronger database constraints.
- ❖ Optimized PostgreSQL queries for search, lease records, and reporting pages by adding indexes, removing N+1 ORM calls, and caching high-read lookup data.
- ❖ Developed asynchronous job workflows with **Celery**, Redis, and retry policies to process document generation, email events, and scheduled customer notifications more reliably.
- ❖ Improved API validation using serializers, schema checks, and reusable error responses, making backend behavior more predictable for React and mobile frontend consumers.
- ❖ Supported migration planning from older Python dependencies to newer stable versions, validating package compatibility through automated regression tests and staging deployments.
- ❖ Collaborated with QA and product teams to reproduce hard-to-detect workflow defects, then delivered targeted backend fixes without introducing broader regression risk.
- ❖ Mentored mid-level developers on **Django**, testing strategy, database modeling, and pragmatic debugging during feature delivery and production support.

Software Engineer

Blackbaud

Jul 2016 – Jan 2018

Charleston, SC

- ❖ Developed backend features for nonprofit fundraising and donor management systems using **Python 3.7-3.8**, **Django**, REST APIs, PostgreSQL, and message-based processing.

- ❖ Improved donor search and reporting workflows by tuning database queries and API response structures, reducing common reporting delays by **21%** for large customer datasets.
- ❖ Implemented secure data import services with validation rules, duplicate detection, and detailed error reporting, reducing manual cleanup effort by **17%** for support teams.
- ❖ Resolved bugs related to malformed CSV imports, timezone conversion issues, and partial transaction failures by adding stricter parsing and rollback-safe processing.
- ❖ Built reusable API components for campaign records, donor profiles, and activity history, helping teams deliver new fundraising features with consistent backend behavior.
- ❖ Supported migration of legacy backend utilities into newer Python modules, replacing fragile scripts with tested services and documented operational procedures.
- ❖ Added automated **Pytest** coverage around critical donation and reporting flows, improving confidence before releases and reducing regression defects.
- ❖ Worked with DevOps to improve deployment checks, environment configuration, and application logging so production issues could be identified faster.
- ❖ Participated in code reviews focused on maintainability, security, and database performance, helping the team keep backend quality stable as product scope expanded.

Software Developer

Granicus

May 2014 – Jun 2016

Dallas, TX

- ❖ Built civic technology backend features using **Python 3.6-3.7**, **Flask**, PostgreSQL, REST APIs, and Linux-based deployment workflows for government-facing platforms.
- ❖ Improved public meeting content processing and notification reliability through better job scheduling, retry handling, and backend validation.
- ❖ Implemented API endpoints for agenda management, document publishing, and subscription workflows while keeping permission checks consistent across internal modules.
- ❖ Fixed recurring production issues involving failed document uploads, missing metadata, and slow search responses by improving file validation and database indexing.
- ❖ Migrated older utility scripts into maintainable Python services with clearer configuration management, logging, and reusable exception-handling patterns.
- ❖ Created integration tests for API workflows that handled public records, meeting documents, and notification delivery, reducing late-cycle defects before deployment.
- ❖ Worked with support teams to investigate customer incidents, reproduce backend errors, and deliver safe patches without disrupting active government portals.
- ❖ Improved developer workflow by documenting setup steps, common troubleshooting commands, and backend release checks for new team members.

Junior Software Developer

Vology

Jun 2013 – Apr 2014

Gainesville, FL

- ❖ Supported internal IT services applications using **Python**, Flask, MySQL, shell scripting, and Linux tools for automation and backend maintenance.
- ❖ Built small automation services for ticket routing, inventory updates, and operational reporting, reducing repetitive manual work for internal teams.

- ❖ Fixed bugs in legacy scripts involving bad input handling, missing null checks, and inconsistent database updates by adding validation and safer error handling.
- ❖ Assisted senior engineers with API maintenance, database scripts, and deployment preparation while learning production debugging and software delivery practices.
- ❖ Wrote unit tests, technical notes, and troubleshooting guides for backend utilities, helping the team maintain older systems with fewer support escalations.

EDUCATION

Arkansas State University

Bachelor's degree, Computer Science (Sep 2009 – May 2013)